Snowline Altitude Estimation in Sentinel-1 Satellite Images Using an Extended Chan–Vese Method: Proof of Concept and Prototype of Fusion With Optical Data

Karlotta Kilias, Guillaume James, Fatima Karbou, Clémence Turbé, Adrien Mauss, and Guillaume Eynard-Bontemps

***Abstract*—Snow cover varies at the scale of a mountain range depending on topography (altitude, orientation, and slope) and weather conditions. We provide a simple synthetic aperture radar (SAR)-based remote sensing tool to calculate snowline altitude estimates (SAEs), i.e., the average elevation above which snow is continuously present. This tool allows to track areas of wet snow and their changes over the seasons, and to monitor the onset and localization of snow melt. Unlike existing approaches, we detect wet snow areas by performing a segmentation on synthesized images in the form of orientation–altitude diagrams. The segmentation is done by means of an enhanced Chan–Vese curve evolution method. Our SAEs are calculated directly from satellite observations, without resorting to existing snow products, and do not fundamentally rely on the choice of thresholds. The algorithm has been tested on Sentinel-1 (S1) images during the melting period of 2018 over a large part of the Northern Alps in France, for which various validation data sets are available. A comparison of the SAEs to the model-based Bulletin d'Estimation du Risque d'Avalanche (BERA) and the optical fractional snow cover copernicus product shows good agreement, with correlations close to 0.86 for the Belledonne massif and above 0.95 for the Vanoise and the Grandes Rousses massifs when comparing SAR-based estimates and BERA. Root mean square errors (RMSEs) were found higher for north-facing slopes than for south-facing slopes. Furthermore, we demonstrate that our algorithm can be directly applied to other types of satellite data, as long as the observations are highly sensitive to snow, by providing an example based on the Sentinel-2 (S2) normalized difference snow index (NDSI). This shows that in the optical case, SAEs can be computed with minimal data, for instance under heavily clouded conditions (depending on the cloud positioning). Finally, we present a proof of concept for a hybrid SAE using a combination of optical and radar data, which represents a crucial progress in the harmonious integration of various types of measurements for snow monitoring.**

***Index Terms*—Meltlines, optical, segmentation, Sentinel-1 (S1), Sentinel-2 (S2), snow, snowlines, synthetic aperture radar (SAR), wet snow.**

I. Introduction

SNOW cover plays a crucial role in Earth's energy balance, as snow reflects solar radiation and insulates the ground below. Moreover, it serves as a protective layer for other cryosphere components, such as glaciers, sea ice and permafrost [1], [2], [3]. There is an overall reduction in low-elevation snow cover attributable to climate change during recent decades with snow-cover duration decreasing in almost all areas, particularly at lower altitudes [4]. This entails rapid changes, especially in vulnerable mountain areas: impacts range from changes in mountain ecosystems (see [5]) to socioeconomic questions like water management and the prevention of natural disasters such as floods, avalanches and rock falls [6], [7]. Since snow-cover decline contributes in turn to the acceleration of global warming, its repercussions extend all over the globe and well beyond mountain areas. See [8] for a very comprehensive review on the current state of the cryosphere in Europe.

Methods for snow cover monitoring provide important data for both short-term prediction of local conditions and long-term understanding of global climate impacts. Measurements can be complicated, as snow is a very dynamic research object with high temporal and spatial variability (see [9]). To capture its often rapid development, observations must be made at a fine temporal scale, and since the areas of interest can be vast, inaccessible, or subject to hostile weather conditions, in situ measurements are often not feasible. Remote sensing techniques are therefore of great value in the field of snow monitoring. For the present study, we use data from the Sentinel-1 (S1) and Sentinel-2 (S2) satellite constellations, operated by the European Space Agency in the frame of the European Earth Observation program Copernicus, with a particular focus on the analysis of S1 Synthetic Aperture Radar (SAR) images. Contrary to optical sensors as carried by S2,

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SAR has the advantage of not being impacted by cloud cover, thus providing exploitable images at a high frequency (six days for the study area in the French Alps). SAR measurements, which are highly sensitive to the presence of liquid water, are limited to the detection of wet snow (i.e., snow containing liquid water). Optical sensors, on the other hand, react to light reflected by wet and dry snow (dry snow consisting only of ice particles and air, with temperatures below 0 ° C) and therefore detect total snow cover. For the present work, in which we focus on melting periods, we make the assumption that there is no dry snow at lower altitudes than wet snow. Under this assumption, SAR acquisitions can provide information on the line between snow and snow-free areas (which we call the snowline) as soon as there is any humidification present in the snow pack. While there are many exceptions due to harsh weather conditions, this important constraint is usually met in the melting season. Note that in this article, we focus exclusively on snowline altitude estimation. However, our algorithm applied to S1 images allows to estimate the altitude of the limit between wet and dry snow (which we call the meltline) at the same time due to SAR's sensitivity to liquid water content.

The complex problem of detecting wet snow areas in high-resolution C-band SAR backscatter images is most often targeted by means of the threshold-based reference method established by Nagler and Rott [10] and Nagler et al. [11]. They use a combination of co- and cross-polarized images (VV and VH respectively, weighted by the local incidence angle), pointing out the added value of cross-polarization in steep topographies [12]. After a preprocessing stage, which includes speckle filtering, the ratio of a given image to a reference image without wet snow is calculated to mitigate terrain influence, and converted to decibel. Subsequently, a threshold is applied to this ratio (−3 dB in the 2000 study, −2 dB in the 2016 study), with values below being classified as wet snow. The threshold is chosen based on the image histograms and has since been adapted to a variety of locations and sensors. Being based on a single acquisition, the algorithm inherently struggles with speckle noise, and Nagler and Rott [10] notice a slight under-detection of wet snow in patchy areas near the snow line. Forest and water are excluded. Robust and easy to apply, Nagler's method has been used in 82% of studies around SAR-based wet snow detection from the past three decades (according to the comprehensive review by [13]).

More recently, other approaches to wet snow detection have been tested, which aim at bypassing the fixed −2.dB threshold, since the latter does not account for variance in surface type or roughness and can introduce detection errors under some circumstances (see [14], [15]). These methods include adaptive thresholds in form of the Otsu method [16], HSV-space segmentation (introduced in [17] and applied to wet snow in [14]), as well as iterative methods, such as the recent dynamical systems approach in [18] and the Chan–Vese curve evolution method ([19], [20]). A number of these approaches are examined in the exploratory study [14], which systematically compares simple segmentation algorithms originating from six different methodological families applied to SAR backscatter ratios. Among other results, they state that all algorithms tend to under-detect snow edges and that methods with a higher degree of smoothing consistently achieve better results according to the scores developed in the study. In this study, the method of Chan–Vese produces the best snow occurrence probability across one month in the melting period when compared with optical products. There are, however, some issues with over-segmentation and class inversions, the computational cost is quite high, and the influence of pixels in geometric distortion zones cannot be eliminated. It is worth mentioning that, although these authors are the first to employ the method of Chan–Vese in the context of snow monitoring to the best of our knowledge, it is regularly applied to other SAR-related research, such as flood mapping (see [21]), ship detection (see [22]) and the monitoring of water bodies (see [23]).

Owing to the recent development of computational equipment and processing capacities, machine learning-based image segmentation algorithms are now powerful options, such as the supervised learning random forest method that has been successfully tested for snow detection on SAR images by Tsai et al. [24], and whose performance has been confirmed in [14]. Note that as [24] incorporates interferometric SAR coherence and polarimetric parameters in addition to the backscattering coefficient, results include not only wet, but also total snow extent, and are applicable to all land cover.

All of the algorithms discussed above segment highly resolved SAR backscatter images directly, thereby creating so-called snow-cover maps. Besides SAR, a wide range of other sensors has been used to estimate snow-cover maps even on steep mountains, notably high and medium resolution optical data ([25], [26], [27], [28], [29], [30], among many other studies).

A very effective way to describe the distribution of snow cover in mountainous environments is to draw a line that separates the areas covered by snow from those that are not. In [31], the snowline altitude is estimated as the elevation for which the sum of snow pixels below and snow-free pixels above that elevation is minimized, based on MODIS snow products. A similar method to track changes in the snowline altitude in the Alps has been investigated in [32]. Snowline elevations in Europe were also determined by [33] and [34] using MODIS and Landsat images, respectively. Other studies have investigated the use of moderate resolution images for monitoring snowline elevations (see [35], [36] among others). Besides the use of satellite data, it has also been demonstrated that terrestrial camera systems are an effective way to monitor changes in the snowline, both at the regional level and at the scale of small catchments [37], [38], [39].

To obtain a snowline altitude estimate (SAE) at the scale of a massif, the segmented image or the binary snow product can be synthesized into an orientation–altitude diagram, where pixels are grouped according to their orientation and altitude information from a digital elevation model (DEM) and represented by a suitable statistical value (see [40]). One target of our study is to provide representations of snow variability in mountainous regions that align with the outputs of the Météo-France snow model chain, the S2M chain [41], [42]. The S2M simulations

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are performed on elementary zones designed specifically to represent the principal factors that influence spatial variability in mountain ranges referred to as "massifs." A mountain range is a conceptual entity that denotes a mountainous area, often covering over 1000 km^2, where meteorological conditions are considered as uniform at a particular elevation. This framework facilitates the representation of complex topography by incorporating the varied elevations, orientations and slopes of a mountain. Indeed, orientation–altitude diagrams provide a comprehensive view of snow distribution and allow for easy calculation of SAEs for different orientations. For any fixed orientation, the snow-cover evolution can subsequently be tracked by means of a time-elevation plot. This diagnostic approach is valuable for applications that emphasize the spatial distribution of snow cover within a particular region, such as a mountain range, rather than traditional snow maps. It can be used for modeling and forecasting snow cover to predict avalanche activity and monitor mountain ecosystems, and allows for an easy comparison of snow products from different sources, regardless of their native resolution.

In the present work, we introduce a novel technique for generating SAEs from satellite data. Compared with the approaches described above, we change the order of segmentation and image synthesis by first aggregating the image information into orientation–altitude diagrams, and then segmenting the latter. This approach minimizes the computational cost while providing sufficient precision at the scale of massifs. For the segmentation, we employ the method of Chan–Vese, which we extend to address some problems that were encountered in the study of [14]. This method allows us to treat data from different sensors in a very similar way. Moreover, the method can be adapted to directly incorporate different input information into the segmentation process, thereby creating hybrid SAEs. We showcase this with an example using SAR and S2 optical images. This work lays the foundation for the development of an efficient snowline altitude product which combines information from all available satellite sources on a given day.

The article is structured as follows. We briefly outline the technical background on S1 SAR and S2 optical acquisitions and describe the data and validation products that we use in Section II. Subsequently, we introduce our approach to calculating SAEs from SAR data in Section III, which consists in applying an extended Chan–Vese segmentation method to synthesized satellite images in an orientation–altitude reference system. We revisit the original implementation of the Chan–Vese method in `scikit-image` and discuss features that were added in this study, in particular for dealing with images with missing data (e.g., geometric distortion in SAR images, clouds in optical images). The SAEs obtained from SAR backscatter ratios are then compared with different validation products in Section IV. Finally, we demonstrate the capability of the algorithm to fuse different data sources by simultaneously segmenting S1 and S2 data in Section V. Section VI is dedicated to the discussion of technicalities and possible improvements of our algorithm, and in Section VII we conclude and give an outlook on future work.

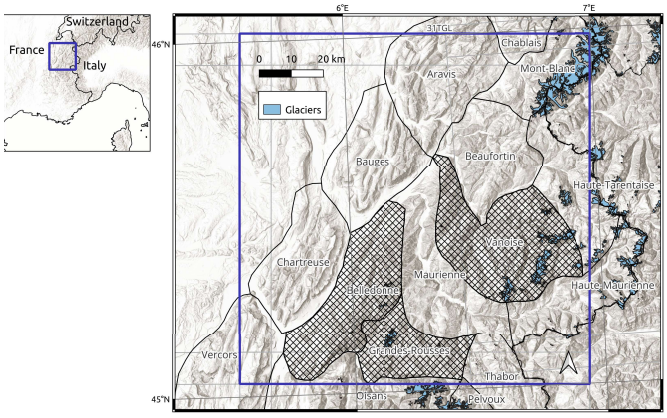


Fig. 1. S1/S2 tile 31TGL following the military grid reference system (MGRS) tiling, with the study area: belledonne, the grandes rousses and the vanoise.

II. Study Data

A. Test Zone and Time Frame

We work on the three massifs of Belledonne, Grandes Rousses and Vanoise in the satellite tile 31TGL in the French Alps. This test zone is selected since it encompasses several massifs for which independent data sets, including in situ measurements, are available. Furthermore, these mountain ranges are sufficiently high (close to or above 3000 m) to monitor snow melt over several months, and present different elevation and vegetation cover characteristics. The massif contours are defined according to the Météo-France standards (see Fig. 1 for an illustration). The temporal frame is the season of 2017/18, which was characterized by abundant snow and a rather late onset of snow melt in early April, with frequent cloud-free conditions during April (see [43]).

B. Data

In the following, we briefly recall some technical properties of S1 and S2 acquisitions to create an understanding of the nature of the data used in this study. Initiated in 2014 and 2015 respectively, the S1 and S2 missions each consist of an identical pair of satellites, carrying radar (S1) or optical (S2) measurement instruments. Images are obtained every six days over Western Europe for S1 (except late 2021–2025, where the revisit time is 12 days due to the loss of S1-B), and every five days for S2. See the Copernicus website `https://sentiwiki.copernicus.eu` for more information on the Copernicus program and the Sentinel missions.

S1 satellites are equipped with SAR. This instrument emits electromagnetic waves at C-band wavelength and measures the backscatter echo from earth, which depends on the surface roughness and dielectric properties (i.e., reflectivity) among other factors. Importantly, owing to the wavelength, SAR detection is neither hindered by cloud coverage nor low light conditions, leading to regular exploitable observations. S1 backscatter images have a high spatial resolution of 20 m due to the electronic simulation of a large antenna ([44]). However, their analysis presents some particular challenges:

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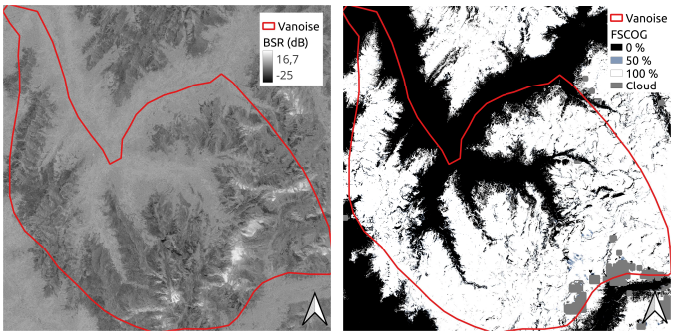


Fig. 2. (Left) SAR BSR on April 16 and (right) the S2 fractional snow-cover product on April 20 in the vanoise.

the side-looking acquisition comes with orbit-dependent geometric distortions in the form of foreshortening, layover and shadow effects. Moreover, phase interference from different ground features creates random speckle noise, which deteriorates the image quality significantly. In order to assess the variation of the backscatter independently of terrain influence, we calculate backscatter (intensity) ratios (BSRs), which correspond to an acquisition divided by a snow-free reference image and converted to dB. Fig. 2 (left) shows an example of an SAR BSR over the Vanoise.

Wet snow monitoring by means of SAR backscatter is based on the fact that the backscatter coefficient σ^0 is lower the more liquid water the snow pack contains due to increased surface scattering and attenuation [12]. Furthermore, the penetration depth into the snow pack is highly dependent on the liquid water content: whereas for wet snow, only the top layer (around 3 cm) contributes to scattering and reflection, dry snow is penetrated up to 20-m deep [13]. This leads to significant ground distributions, so that contrasts between dry snow and snow-free conditions are too small to discriminate between them. Although total (including dry snow) detection is possible using interferometric and polarimetric SAR (see [24]), SAR backscatter, as used in the present study, is therefore suitable only for wet snow detection.

For the SAR data, we use open access level-1 ground range detected products acquired in Interferometric Wide swath mode under cross-polarization (see [11]). They have been preprocessed with the S1 Tiling Tool (`https://github.com/CNES/S1Tiling`), which performs data download, calibration, speckle filtering and orthorectification [45]. The preprocessed images are orthorectified to the S2 grid and have a pixel size of 10 m. Computations have been done using acquisitions from the ascending orbit relative number 161 (taken around 17 h 30 min) and the descending orbit relative number 139 (taken around 5 h 45 min). However, for simplicity, we consider only the ascending orbit in this article, where we expect more wet snow due to the time of day. As reference image, we use a four-year summer average (2020–2023, from July to September respectively).

S2 satellites are equipped with optical sensors that cover 13 spectral bands (443–2189-nm wavelength). Resolutions range from 10 m for the visible to 20 m for the red edge and 60 m for the near infrared bands. Optical sensors detect sunlight reflected by the Earth, and therefore do not discern between dry and wet snow; they are used to deduce total snow cover. Unlike SAR, S2 is strongly affected by clouds, cloud shadows and (polar) darkness, which introduces a risk of confusion (between snow, ice, and clouds) and can render acquisitions completely unexploitable. Note that these conditions are all the more recurrent given the seasons and regions of interest. The S2 reflectance products were downloaded from the Theia website `https://theia.cnes.fr`

For the transfer to the orientation–altitude reference system, we make use of the 30-m SRTM DEM, which has been oversampled to 10 m for pixel co-localization.

TABLE I

CORRELATIONS AND RMSE (IN M) BETWEEN SAR SAES AND THE DIFFERENT VALIDATION DATASETS FOR SOUTHERN ORIENTATIONS

SOUTH	Belledonne		Grandes Rousses		Vanoise	
Type	Corr.	RMSE	Corr.	RMSE	Corr.	RMSE
SAR/FSC	0.86	233.1	0.94	235.7	0.98	178.5
SAR/BERA	0.91	236	0.95	239.3	0.98	198.4
FSC/BERA	0.93	156	0.94	162.5	0.98	96.4

TABLE II

CORRELATIONS AND RMSE (IN M) BETWEEN SAR SAES AND THE DIFFERENT VALIDATION DATASETS FOR NORTHERN ORIENTATIONS

NORTH	Belledonne		Grandes Rousses		Vanoise	
Type	Corr.	RMSE	Corr.	RMSE	Corr.	RMSE
SAR/FSC	0.8	216.7	0.95	294.2	0.92	283.1
SAR/BERA	0.86	364.7	0.96	415.3	0.95	429.9
FSC/BERA	0.83	333.9	0.95	204.5	0.95	239.3

C. Validation Products

As optical reference product, we employ the S2 Fractional Snow Cover On Ground (FSC) product downloaded from the Copernicus Cryoland GeoPortal (`https://cryo.land.copernicus.eu`). Calculated from the normalized difference snow index (NDSI) by means of a specifically developed function ([46]), the FSC indicates the fraction of each pixel covered by snow with a resolution of 20 m. An illustration can be found in Fig. 2 (right). To generate a data set that can be compared with the seasonal SAR SAEs, the FSC has been transferred to the orientation–altitude system (see Section III) by taking the arithmetic mean of each class. A segmentation is then obtained through the application of a threshold at 50% average pixel cover. We perform the snowline altitude detection exactly as for the SAR images (see Section III-C). We used only FSC images where at least 1% of pixels were exploitable (i.e., not in a valley and cloud-free). In general, the FSC algorithm could be improved, as the segmentations obtained vary a lot in quality and are often very pixelized. The FSC snowline altitudes are therefore to be interpreted as an approximate reference.

Moreover, we refer to the Bulletin d'Estimation du Risque d'Avalanche (BERA), which is based on snow models, different (in situ) observations and human expertise. The BERA is provided daily during the winter season by the Météo-France weather and avalanche risk forecasters and includes a range of different information such as snowline altitudes and rain-snow limits. For the purpose of this work, we use the snowline

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TABLE III

CLOUD PERCENTAGES FOR THE DATES DISPLAYED IN FIGS. 5 AND 6 (SOUTH: 337.5-22.5°, NORTH: 157.5-202.5°). WHEN THERE IS NO ACQUISITION FOR THE GRANDES ROUSSES AND THE VANOISE, THE ACQUISITION FOR BELLEDONNE WILL ALSO BE ONLY PARTIAL. WE INDICATE THE CLOUD PERCENTAGE IN THE VISIBLE PART

	Belledonne		Grandes Rousses		Vanoise	
Date	South	North	South	North	South	North
03/04	46	40	-	-	-	-
03/14	62	60	-	-	-	-
03/16	31	24	18	14	55	47
03/21	94	94	68	63	85	68
03/24	37	32	-	-	-	-
03/26	98	99	85	87	92	92
03/31	91	91	35	36	92	87
04/05	91	95	48	45	82	80
04/10	66	65	38	43	100	100
04/15	12	11	1	1	53	39
04/18	42	37	-	-	-	-
04/20	0	0	0	0	6	3
04/25	11	11	1	1	1	3
04/28	82	73	-	-	-	-
04/30	91	89	86	82	98	96
05/05	10	12	44	38	85	82
05/08	99	99	-	-	-	-
05/18	88	86	-	-	-	-
05/20	84	83	100	100	100	100
05/25	12	13	28	26	24	26
05/28	75	69	-	-	-	-
06/02	91	86	-	-	-	-
06/09	64	66	81	80	92	93

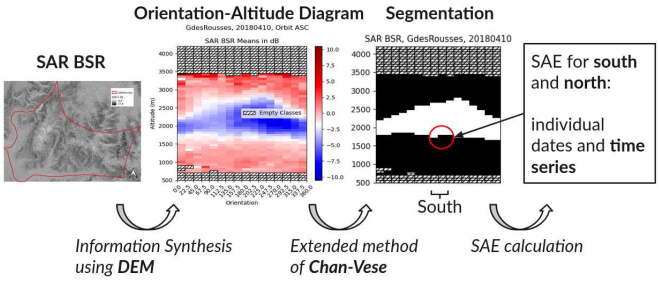


Fig. 3. Schematic view of the snowline estimation algorithm.

altitude information for southern and northern orientations. When no snowline information is accessible on a given date, the last available snowline altitude is indicated in the BERA, creating a step-like pattern.

III. ALGORITHM

Our algorithm is presented as follows. In Section III-A, we outline the transformation of SAR BSR images to orientation–altitude diagrams. Section III-B introduces the method of Chan–Vese with our extension, which we employ to segment the orientation–altitude diagrams. In Section III-C, we explain how to infer the SAE from the segmentation, and Section III-D briefly comments on computation time. A schematic overview of the workflow can be found in Fig. 3.

A. Transformation to the Orientation–Altitude System

For each massif, we calculate the BSR by dividing the SAR backscatter image through a four-year summer reference (to mitigate terrain influence) and converting the result to dB (by applying $10\log_{10}$). The BSR is then synthesized into an

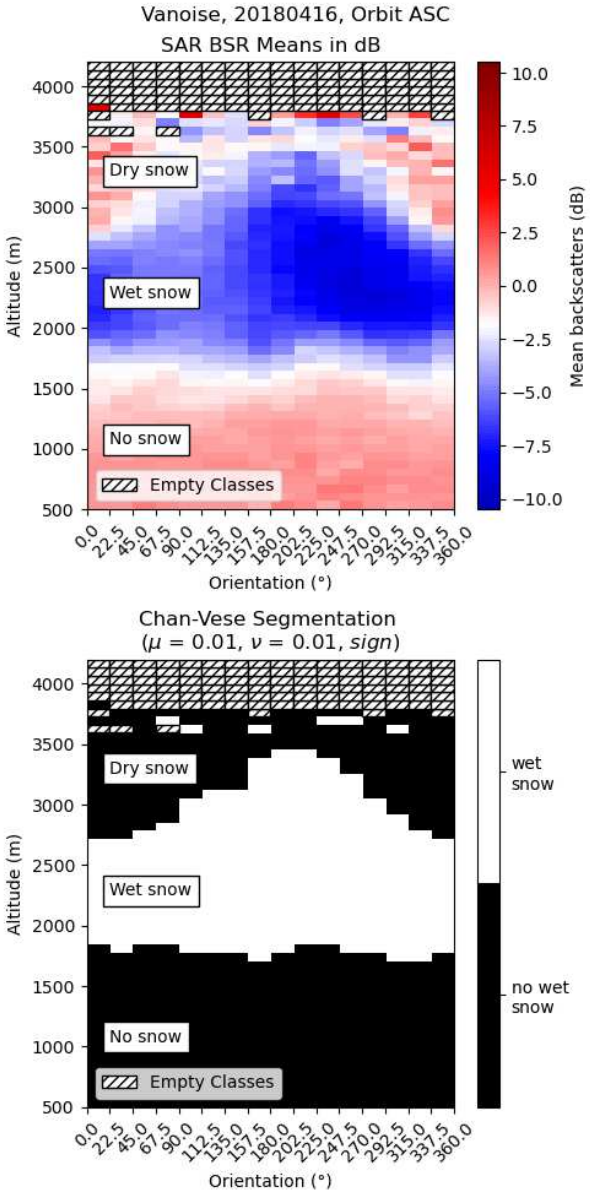


Fig. 4. (Top) SAR BSR arithmetic means (in the logarithmic domain) in the vanoise on April 16, 2018. (Bottom) Corresponding Chan–Vese segmentation (before morphological filtering).

orientation–altitude diagram using the orientation and altitude information from the DEM. The orientation–altitude diagram corresponding to the BSR depicted in Fig. 2 is shown in Fig. 4 (top). Geometric distortion zones as well as glaciers are set to NaN ("not-a-number") beforehand, and only moderate slopes (10° – 30°) are considered for better comparability with the BERA. We represent each class by the arithmetic mean of the contained pixels taken in the logarithmic domain (i.e., after conversion to dB). This corresponds to taking the geometric mean in the linear domain, which aligns with an interpretation of the BSR as a multiplicative entity (a ratio). Other options that were tested are the class medians, which produced similar results, and the arithmetic mean in the linear domain, which led to higher BSR values (the arithmetic mean of a set of numbers always being larger or equal to the

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geometric mean). A look at the in-class distributions confirms that both arithmetic mean in the logarithmic domain and median are good options, since we deal with approximate log-normal distributions. Note that the choice of representative is of great importance especially around the snowline, and has to be adapted to the respective data type.

The orientation–altitude coordinate system has been employed in [40] to analyze wet snow properties at the scale of massifs. In our case, the resulting aggregated image is segmented with an extended version of the method of Chan–Vese as described in Section III-B.

B. Extension of the Method of Chan–Vese

To estimate SAEs from the orientation–altitude diagrams, we first segment the latter using the method of Chan–Vese [20], [47]. In this section we describe the original method and an alteration that we made in order for it to be applicable in our case.

As a so-called *Active Contour Model*, the method of Chan–Vese aims at detecting object contours in (most often) greyscale images by making a curve C evolve on the image domain so that an image-dependent energy is being minimized. In the case of Chan–Vese, one obtains a segmentation of the image into two constant phases corresponding to the inside and outside regions of C. Let $f : \Omega \subset \mathbb{R}^2 \to [0, 1]$ an image, and define an initial region $\Omega_0 \subset \Omega$ such that $\delta\Omega_0 = C$. The Chan–Vese minimization problem reads as

$$\begin{aligned} &\underset{c_1,c_2,C}{\arg\min} \ \mu \text{Length}\,(C) + \nu \text{Area}\,(\text{int}\,(C)) \\ &+ \lambda_1 \int_{\text{int}(C)} |f\,(x) - c_1|^2\,dx + \lambda_2 \int_{\text{ext}(C)} |f\,(x) - c_2|^2\,dx, \\ &\mu \geq 0, \lambda_1, \lambda_2 > 0, \quad \nu \in \mathbb{R} \end{aligned} \tag{1}$$

with the curve-dependent constants $c_{1,2} \in [0, 1]$. $\lambda_{1,2}$ control the homogeneity of the regions inside and outside of C, μ penalizes the edge length, i.e., smoothness of C, and ν allows to penalize the surface of the detected area (positively or negatively).

To solve problem (1), Chan and Vese employ the level set method by Osher and Sethian [48], which has the benefit of allowing for topological changes and the creation of non-smooth features. Let $\varphi : \Omega \to \mathbb{R}$ a smooth level set function, i.e., a smooth function such that

$$\begin{cases} \varphi\,(x) > 0, & x \in \Omega_0 \\ \varphi\,(x) < 0, & x \in \overline{\Omega}_0^c \\ \varphi\,(x) = 0, & x \in \delta\Omega_0 = C. \end{cases}$$

The energy term in (1) can be expressed as a function of $\varphi, c_{1,2}$ and optimized for φ and $c_{1,2}$ separately. We present only the results and refer to [20] and [47] for details. Optimization with respect to φ is performed by means of the following gradient flow, which is directly derived from the energy term through functional derivation with respect to φ

$$\begin{cases} \dfrac{\partial \varphi}{\partial t} = \delta_\varepsilon\,(\varphi) \left(\mu \text{div} \left(\dfrac{\nabla \varphi}{|\nabla \varphi|} \right) - \nu - \lambda_1\,(f - c_1)^2 \right. \\ \qquad\qquad \left. + \lambda_2\,(f - c_2)^2 \right) \\ \varphi\,(x, y, 0) = \varphi_0\,(x, y) \quad \forall\ (x, y) \in \Omega \\ \dfrac{\delta_\varepsilon\,(\varphi)}{|\nabla \varphi|} \dfrac{\partial \varphi}{\partial n} = 0 \quad \forall\ (x, y) \in \delta\Omega \end{cases} \tag{2}$$

where $\delta_\varepsilon(t) = (\varepsilon^2/\pi(\varepsilon^2 + t^2))$ is a regularized Dirac, and φ_0 defines an initial level set function (see Section III-B for details on the initial condition). The boundary conditions are a consequence of the Green–Ostrogradski theorem. At each iteration, the φ−dependent values $c_{1,2}$ are recalculated as $c_1 = \text{mean}(f_{|\Omega_0})$, $c_2 = \text{mean}(f_{|\overline{\Omega}_0^c})$, resulting from the necessary conditions for an extremal point of the energy for fixed φ. Subsequently, φ is updated according to the gradient flow (2). For the spatial discretization of the curvature term $\text{div}((\nabla\varphi/|\nabla\varphi|))$, Chan and Vese employ a mixture of forward and backward finite difference stencils. In time, a semi-implicit Gauss–Seidel method is used, which computes the value at each grid point individually (bottom-to-top, right-to-left), considering the latest available value of all its neighbors.

The present work extends an existing implementation of the method of Chan–Vese from the Python-library `scikit-image`. Rather than using the Gauss–Seidel method from the original paper, the developers use a Jacobi-method, where all values used for the calculation stem from the previous iteration alone. Although this algorithm converges more slowly, the implementation is ultimately more efficient with a Jacobi-method, as it can be performed using only array operations. For a fixed rectangular grid with points (x_j, y_j) so that $1 \leq i, j \leq M$ and time step size Δt we obtain the following discretization for time step $n + 1$:

$$\begin{aligned} \varphi_{i,j}^{n+1} = &\left(\varphi_{i,j}^n + \Delta t \cdot \delta_\varepsilon \left(\varphi_{i,j}^n \right) \left(A_{i,j}\varphi_{i+1,j}^n + A_{i-1,j}\varphi_{i-1,j}^n \right. \right. \\ &+ B_{i,j}\varphi_{i,j+1}^n + B_{i,j-1}\varphi_{i,j-1}^n \\ &\left. \left. -\nu - \lambda_1 \left(f_{i,j} - c_1 \right)^2 + \lambda_2 \left(f_{i,j} - c_2 \right)^2 \right) \right) \Big/ \\ &\left(1 + \Delta t \cdot \delta_\varepsilon \left(\varphi_{i,j}^n \right) \right) \left(A_{i,j} + A_{i-1,j} + B_{i,j} + B_{i,j-1} \right) \\ A_{i,j} = &\frac{\mu}{\sqrt{\eta^2 + \left(\varphi_{i+1,j}^n - \varphi_{i,j}^n \right)^2 + \left(\left(\varphi_{i,j+1}^n - \varphi_{i,j-1}^n \right)/2 \right)^2}} \\ B_{i,j} = &\frac{\mu}{\sqrt{\eta^2 + \left(\left(\varphi_{i+1,j}^n - \varphi_{i-1,j}^n \right)/2 \right)^2 + \left(\varphi_{i,j+1}^n - \varphi_{i,j}^n \right)^2}} \end{aligned} \tag{3}$$

where $\eta > 0$ is a small constant (10^{-5} in our case) added to avoid zero division. The homogeneous Neumann boundary conditions at the image edges [the second equation in (2)] are implemented through one layer of pixel duplication. Finally, the algorithm stops when a predefined number of iterations is reached or the L^2-distance between consecutive iterations of φ falls below a user-defined threshold.

The method of Chan–Vese as provided in `scikit-image` cannot be directly applied to the orientation–altitude diagrams, as the latter contain NaN values for empty classes, which

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expand throughout the iterations. This expansion follows naturally from the scheme (3): if $\varphi^n_{i+1,j}$, $\varphi^n_{i-1,j}$, $\varphi^n_{i,j+1}$ or $\varphi^n_{i,j-1}$ are NaN, $\varphi^{n+1}_{i,j}$ will be NaN as well; any NaN values are thus transmitted to the neighboring grid points and eventually dominate the whole image.

To avoid this problem, NaN's could be interpolated in a preprocessing step. However, interpolation becomes very approximate when the missing areas are large, and the choice of the interpolation strategy always has an impact on the overall segmentation. For these reasons, we extend the method by implementing homogeneous Neumann boundary conditions at the edges of NaN zones instead of prefilling them, which equally prevents NaN's from spreading. The implementation of the boundary conditions is described in the following. Note that all functions that are sensitive to NaN values have also been modified (e.g., by replacing `numpy.min` by `numpy.nanmin`).

The scheme from (3) can be rearranged to

$$\begin{aligned}\varphi^{n+1}_{i,j} = \varphi^n_{i,j} + \Delta t\ \delta_\varepsilon\left(\varphi^n_{i,j}\right)\Big(&A_{i,j}\left(\varphi^n_{i+1,j} - \varphi^{n+1}_{i,j}\right) \\ &+ A_{i-1,j}\left(\varphi^n_{i-1,j} - \varphi^{n+1}_{i,j}\right) \\ &+ B_{i,j}\left(\varphi^n_{i,j+1} - \varphi^{n+1}_{i,j}\right) + B_{i,j-1}\left(\varphi^n_{i,j-1} - \varphi^{n+1}_{i,j}\right) \\ &- \nu - \lambda_1\left(f_{i,j} - c_1\right)^2 + \lambda_2\left(f_{i,j} - c_2\right)^2\Big).\end{aligned}$$

In $A_{i,j}$ and $B_{i,j}$, we reformulate

$$\begin{aligned}\varphi^n_{i,j+1} - \varphi^n_{i,j-1} &= \left(\varphi^n_{i,j+1} - \varphi^n_{i,j}\right) - \left(\varphi^n_{i,j-1} - \varphi^n_{i,j}\right) \\ \varphi^n_{i+1,j} - \varphi^n_{i-1,j} &= \left(\varphi^n_{i+1,j} - \varphi^n_{i,j}\right) - \left(\varphi^n_{i-1,j} - \varphi^n_{i,j}\right).\end{aligned} \tag{4}$$

We see that if $\varphi^n_{i,j}$ is NaN, $\varphi^{n+1}_{i,j}$ is also NaN, so that all initial NaN's are preserved throughout the iterations. We now focus on stopping the propagation. When computing the iteration φ^{n+1}, the finite differences in the calculation of $A_{i,j}$ and $B_{i,j}$ [expressed as in (4)] are stored in designated matrices. We apply the function `numpy.nan_to_num` to these matrices to set any NaN-containing differences to zero, thereby enforcing Neumann conditions. Since this modification only concerns the matrices storing the differences and does not actually modify any values of φ, we also set $A_{i,j}$ to 0 when $\varphi^n_{i+1,j}$ is NaN, and in doing so, impose $\varphi^n_{i+1,j} = \varphi^{n+1}_{i,j} \approx \varphi^n_{i,j}$ in an efficient manner (equivalently for the other parts of the sum, $A_{i-1,j}$ with $\varphi^n_{i-1,j}$ etc.) Note that with this modification, we make a slight change in the choice of implicit and explicit terms. Finally, in the matrix φ^n, we can set all NaN's to 0, since they have been accounted for.

To illustrate this approach, assume that for fixed i, j the values $\varphi^n_{i,j}, \varphi^n_{i,j+1}, \varphi^n_{i-1,j}$, and $\varphi^n_{i,j-1}$ are non-NaN, but $\varphi^n_{i+1,j}$ is. In the original implementation, $\varphi^{n+1}_{i,j}$, will be NaN. With the modification, $A_{i,j} = 0$ and $\varphi^n_{i+1,j} = \varphi_{i,j}$ in $B_{i,j}$. $\varphi^n_{i+1,j}$ no longer appears on the right side and $\varphi^{n+1}_{i,j}$ is not NaN.

Due to its fast convergence, the standard initial condition φ_0 for the algorithm is the checkerboard shape. However, this initialization comes with the disadvantage that the extent of the detected area cannot be penalized to correct biases (i.e., $\nu = 0$), as we do not know which one of the two regions is being penalized. To amend this issue, we create the image-dependent initial condition $sign_{-2}$, which assigns each pixel to one of two classes depending on whether its value lies above or below −2 dB, and subsequently smoothens the result. With $sign_{-2}$, regions classified as "wet snow" in the beginning will most likely evolve into the regions classified as "wet snow" in the end, allowing us to penalize them. Note that for the same reason, segmentations initialized with $sign_{-2}$ do not have to be oriented in the end, while checkerboard segmentations do. Throughout the melting period (roughly April to June), the two initial conditions lead to very similar results, but significant differences tend to appear before and after. Other initial conditions (notably not SAR-specific ones) are possible as well and some have been tested; for more comments on the initialization, we refer to Section VI.

For our tests, we use $sign_{-2}$ and set $\mu = 0.01, \nu = 0.01, \lambda_1 = \lambda_2 = 1, \Delta t = 0.5$ and the maximum number of iterations to 1000. An example for the segmentation of an orientation–altitude diagram with the extended method of Chan–Vese is shown in Fig. 4 (bottom).

C. Snowline Altitude Estimation

Once the orientation–altitude diagram is segmented, the SAE can be inferred in a straightforward manner. To mitigate the impact of single noise pixels (as observed in Fig. 4) on the detection, we first apply a morphological closing followed by an opening to the segmentation. This removes individual white pixels on black background and vice versa. Starting from the bottom of the segmentation at a fixed orientation, we subsequently identify the first wet snow pixel in upward direction, whose altitude indicates the SAE for this orientation. Empty classes (with value NaN), are skipped in this process. Depending on the desired precision with respect to the orientation, the SAE can be calculated as mean altitude between several of the 16 orientation classes. In our case, southern orientations range from 337.5° to 22.5°, and northern orientations from 157.5° to 202.5°; therefore both southern and northern snowline elevations are averages from two orientation classes. Note that meltline altitude detection can be easily included in this step by continuing upward from the snowline until reaching the first pixel that indicates the absence of wet snow.

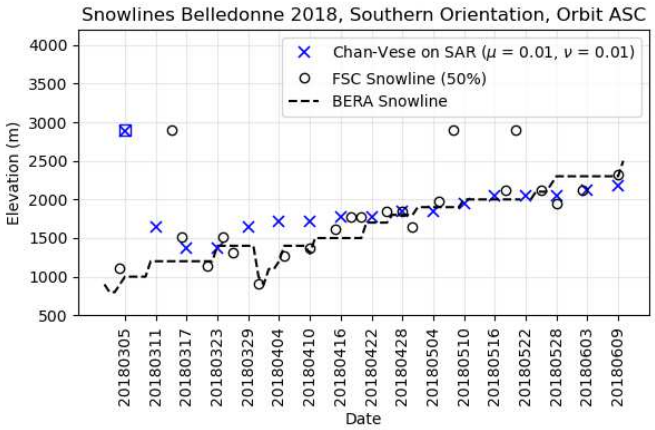
D. Computation Time

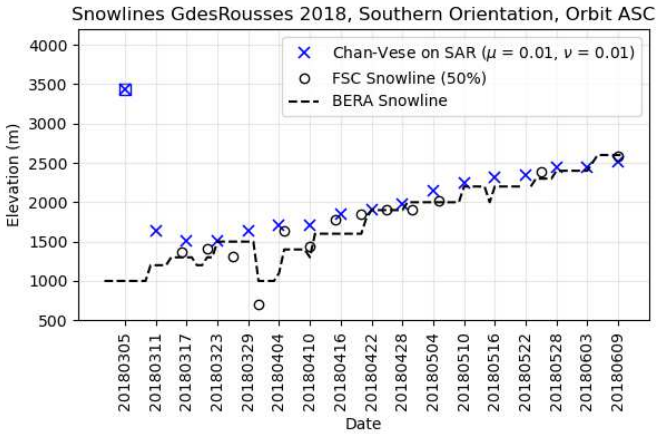
The total computation time from zero for the Grandes Rousses is 25 s on four cores (eight threads) of 2.80 GHz, 32-GB RAM memory. This includes realigning and cutting the image of the whole satellite tile, the reference image and the DEM information as well as the glacier and layover files. To avoid recomputations, all intermediary information is stored, and a recomputation depth can be indicated by the user. The most costly computation is the transformation to the orientation–altitude system, which has been parallelized using the `joblib` library to run on all available cores, reducing its computation time from 55 to 15 s. The segmentation takes about 2 s.

IV. Results for SAR Data

In this section, we present the SAEs obtained from March to June 2018 for the massifs of Belledonne, Grandes

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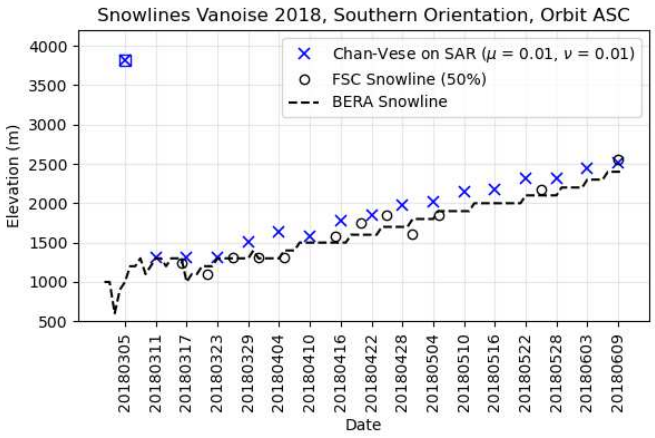
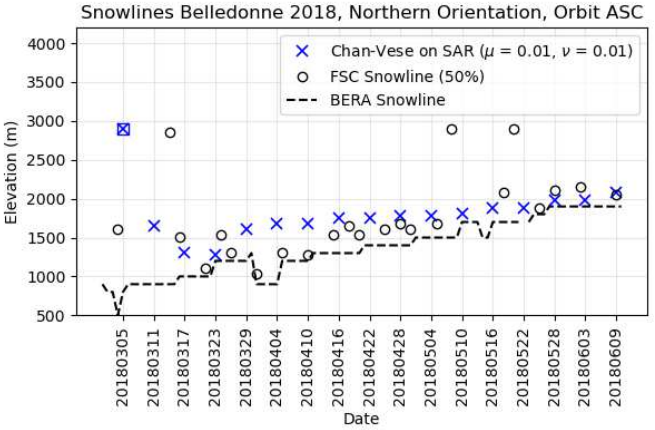
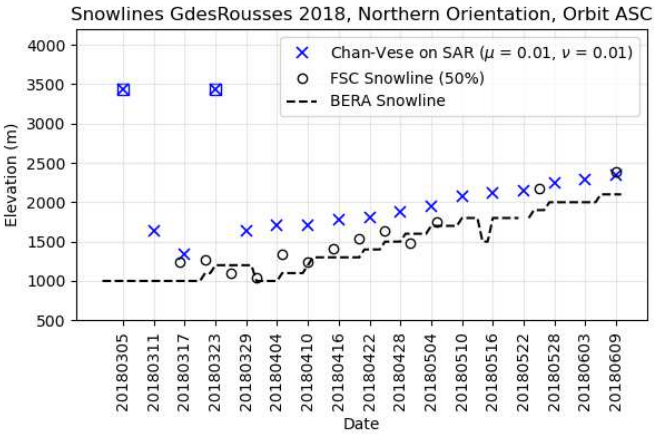


Fig. 5. SAEs for the three massifs in southern orientation. A cross surrounded by a square indicates that no wet snow was detected.





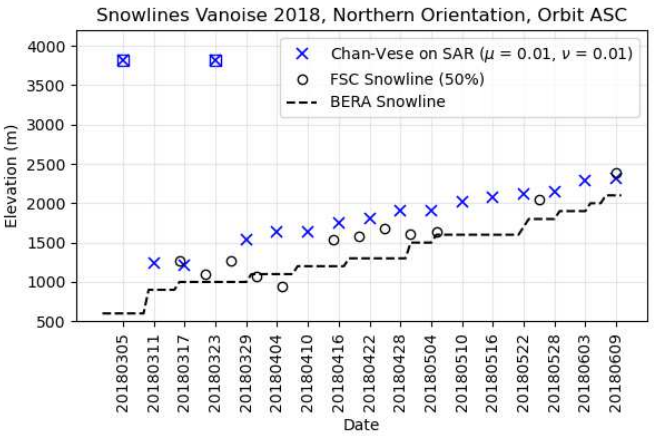


Fig. 6. SAEs for the three massifs in northern orientation. A cross surrounded by a square indicates that no wet snow was detected.

Rousses and Vanoise in southern and northern orientation. Figs. 5 and 6 show the SAEs along with the BERA snowline altitudes as well as the snowline altitudes deduced from the (optical) FSC product as explained in Section II-C. To quantify the quality of our results, we provide correlation and root mean square error (RMSE) scores in Tables I and II. Note that while the correlation results are relatively straightforward to interpret, there remains an open question with regards to acceptable RMSE values. The model outputs that are taken into account for the BERA have altitude intervals of 300 m. For the discussion, we also refer to the segmentation results for the Vanoise depicted in Fig. 7, which illustrate the rising of the snowline and the general humidification of the snow cover during the melting period.

In southern orientation, we generally see a good correspondence between the two validation products FSC and BERA, with correlations upward of 0.93 and RMSEs from 96 to 162 m. In Belledonne, all three FSC outliers were excluded from the scores, as they correspond to partial acquisitions where only pixels at low altitude were admissible due to cloud cover. The SAR SAEs are situated slightly above both validation products, with RMSEs below 200 m for the Vanoise and 230–240 m for the other massifs. The gap to the FSC is coherent with the fact that SAR detection typically struggles with patchy snow around the snowline (see [11]), which we

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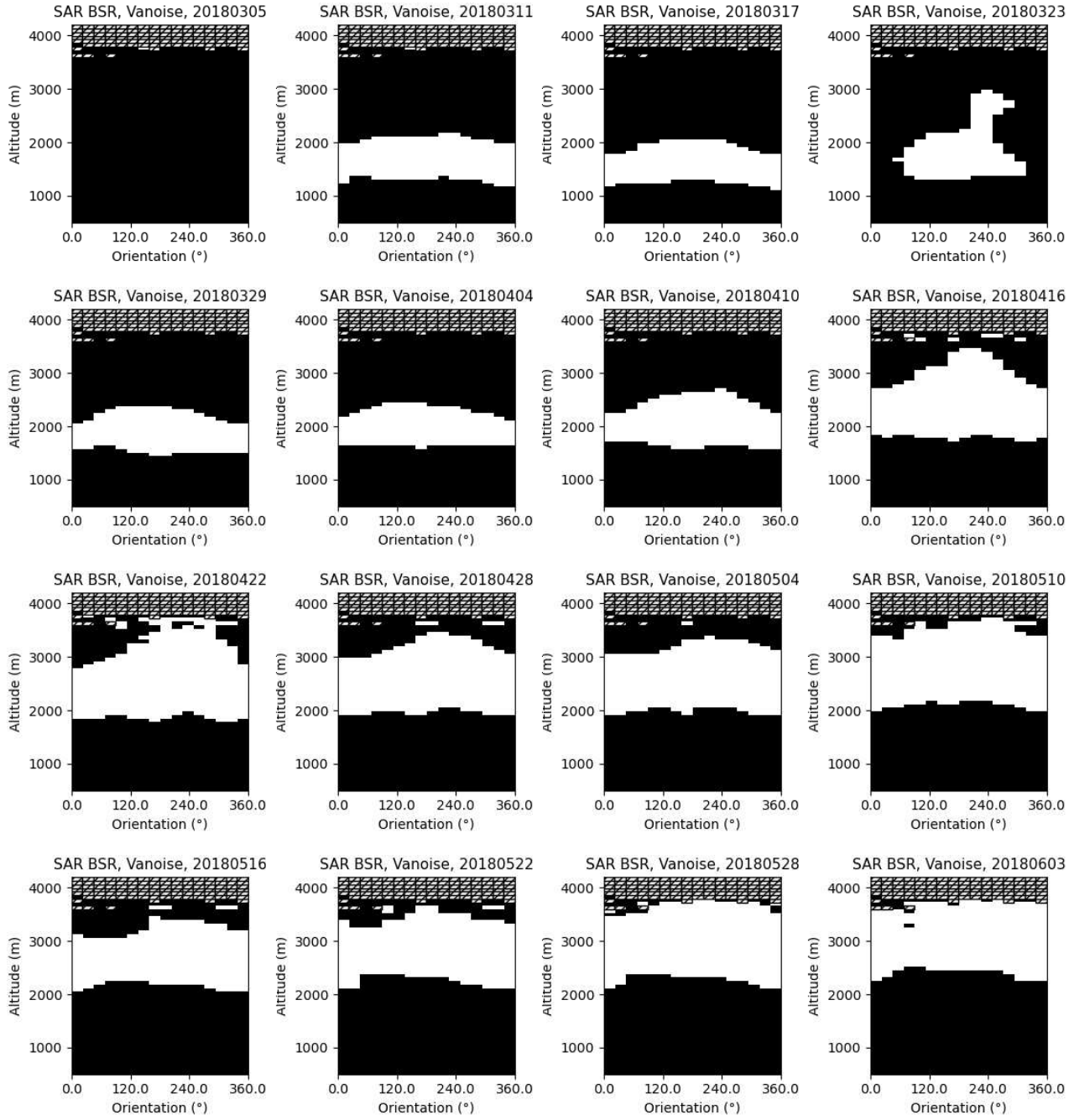


Fig. 7. SAR BSR segmentations throughout the melting period of 2018 in the vanoise. We initialized with $sign_{-2}$ and set $\mu = \nu = 0.01$. White pixels indicate the presence of wet snow, black pixels its absence, and hatched pixels correspond to empty classes.

do not necessarily expect from optical sensors. Correlation scores are uniformly high (0.86 in Belledonne and 0.91–0.98 elsewhere). In Belledonne and the Grandes Rousses, the SAR method did not detect the low altitude snow in late March (no SAR image was taken on the same day). On March 5, no wet snow was detected in any of the massifs (see the segmentation in Fig. 7).

For the northern orientations as presented in Fig. 6, results look quite different. We remark that the FSC product tends to be situated significantly higher above the BERA (RMSEs between 205 and 334 m), although correlation scores remain high, especially for the Grandes Rousses and the Vanoise. As expected, the SAR estimates again lie above the FSC. While RMSEs between SAR and FSC remain somewhat comparable to the southern orientation (216–294 m), RSMEs between SAR and BERA now reach 365–430 m. These large discrepancies to the BERA in northern orientations, which have also been observed in [18], need further investigation. Some elements of a response could be found in the vegetation distribution, given that the gaps tend to get smaller throughout the season: 75% of the forest in Belledonne are located below 1300 m, and below 1700 m for the Grandes Rousses and the Vanoise.

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Finally, we observe that no wet snow was detected in northern orientation on March 23 in both the Grandes Rousses and the Vanoise. Fig. 7 shows a localized accumulation of wet snow in all orientations but the north on this day. In accordance with the BERA report from March 23, we assume that while the wet snow that had frozen overnight stayed frozen in northern orientations, the sunny conditions led to a strong humidification elsewhere.

V. Toward Data Fusion

Since it does not essentially depend on any given threshold, our algorithm can be extended to different data sources in a straightforward manner. In this part of this article, we explain how to transfer the procedure to optical sensors and showcase the possibility of merging SAR and optical image information using the extended method of Chan–Vese. We use S2 data for demonstration purposes.

When dealing with S2 acquisitions, we face the issue of cloud cover, which we handle by setting cloud pixels to NaN before the transfer to the orientation–altitude reference system (using the cloud mask provided with the reflectance images). An overview of cloud percentages for the different orientations can be found in Table III. The object commonly used for snow detection from optical images in the literature is the NDSI introduced in [49]. This index between -1 and 1 corresponds to the normalized difference between band 3 in the green visible spectrum and band 11 in the short-wave infrared

$$\mathrm{NDSI} = \frac{\mathrm{B3} - \mathrm{B11}}{\mathrm{B3} + \mathrm{B11}}.$$

While [25] additionally employ a threshold on the red reflectance to avoid false snow detections on lakes or rivers, we restrict ourselves to the segmentation of the NDSI. Furthermore, we consider only pixels with $\mathrm{B3} > 0$ and $\mathrm{B11} \geq 0$. Unlike in SAR BSR images, the values in any given class of the NDSI orientation–altitude diagrams are usually not log-normally distributed, but often exhibit one or two peaks close to the values -1 and 1. This entails more information loss when representing the classes by a single value. Again, we choose the arithmetic mean as class representative, since it leads to smoother orientation–altitude diagrams than the median. Finally, we modify the initial condition $sign_{-2}$ to $sign_0$ (again, for different options, see Section VI) and negate the orientation–altitude diagram before segmentation, so that the snow region corresponds to the lower values as it does for SAR. Apart from these alterations, the algorithm is applied exactly as in Section III.

An example of NDSI segmentation for June 6 in Belledonne is shown in Fig. 8. That day, only 11% of the pixels over Belledonne were free of clouds, but the algorithm still enables us to deduce snowline altitude information. The quality of the SAE depends directly on the quantity and localization of cloud-free pixels in the study area, and offers only an indication when clouds are too numerous or badly positioned (i.e., covering the snowline).

If for a given date both SAR and S2 acquisitions are available, we can extend the minimization problem to incorporate the variances of both images, thereby obtaining a hybrid

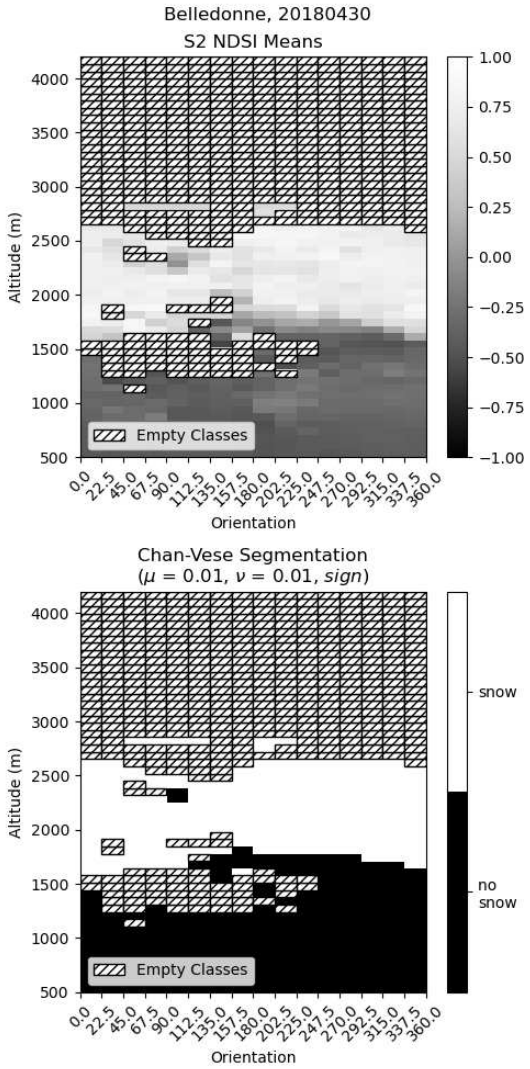


Fig. 8. (Top) NDSI arithmetic mean values in belledonne on April 30, 2018, where there are only 11% of cloud-free pixels in the massif. (Bottom) Corresponding Chan–Vese segmentation (before morphological filtering).

SAE. This is possible as the orientation–altitude reference system is independent of the image input resolution. Noting the (normalized) SAR BSR means as f and the (normalized) S2 NDSI means as g, the energy term then reads as

$$\begin{aligned}&\arg\min_{c_1,c_2,C} \ \mu\mathrm{Length}\,(C) + \nu\mathrm{Area}\,(\mathrm{int}\,(C))\\&+\alpha\left(\lambda_1\int_{\mathrm{int}(C)}|f(x)-c_1|^2dx+\lambda_2\int_{\mathrm{ext}(C)}|f(x)-c_2|^2dx\right)\\&+\beta\left(\lambda_3\int_{\mathrm{int}(C)}|g(x)-c_3|^2dx+\lambda_4\int_{\mathrm{ext}(C)}|g(x)-c_4|^2dx\right)\\&\mu\geq 0,\lambda_{1,2,3,4}>0,\nu\in\mathbb{R},\alpha,\beta\geq 0,\alpha+\beta=1\end{aligned}$$

where α and β are used to tune the degree to which the minimization is driven by each of the inputs: for instance, for $\alpha = 1, \beta = 0$, we obtain the segmentation of the SAR BSR. Since both images are normalized in the Chan–Vese algorithm, their variances are of the same order of magnitude, and the images have equal influence on the minimization if we set $\alpha = 0.5, \beta = 0.5$. The parameters allow for flexibility, as we can assign a smaller weight to the optical input in

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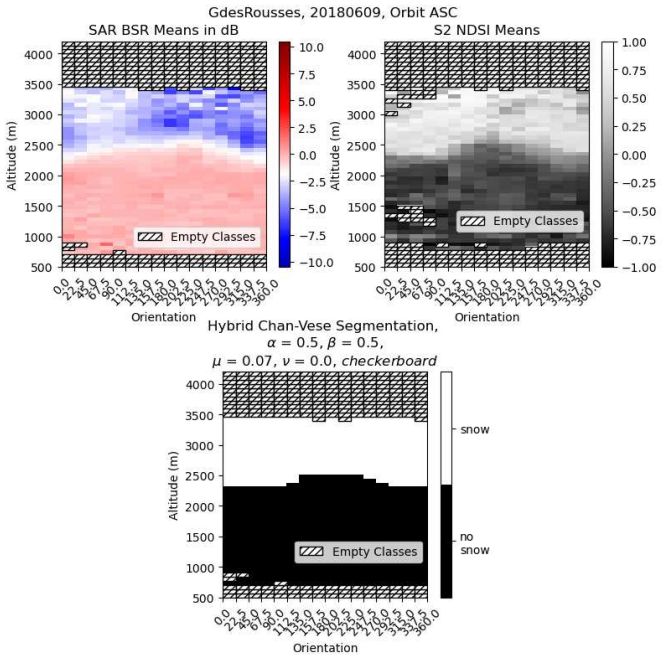


Fig. 9. (Top) BSR and NDSI arithmetic mean values in the grandes rousses on June 9, 2018. (Bottom) Hybrid Chan–Vese segmentation with equal weighting ($\alpha = 0.5, \beta = 0.5$) and checkerboard initialization.

the case of significant cloud cover as in Fig. 8. Note also that the coefficients where exactly one of the two source images is NaN are replaced with the respective region means, so that the resulting segmentation is NaN only where both images are NaN. For hybrid segmentations, we use the image-independent initial condition checkerboard with the parameters $\mu = 0.07, \nu = 0, \lambda_{1,2,3,4} = 1$ (as explained in Section III-B, the area cannot be penalized in this case). Fig. 9 shows an example of a hybrid segmentation with equal weights from June 9. Note that for this date, the individual segmentation of the SAR BSR will not produce a meltline, indicating that all existing snow is wet, so that a hybrid segmentation is pertinent.

VI. Discussion

This section is dedicated to some choices we made during the development of the algorithm as well as a few remaining unresolved issues.

With respect to the initial condition for nonhybrid segmentations, we would like to point out that it is possible to generalize $sign_{-2}$ and $sign_0$ to different data types by choosing the mean value of the image as threshold. This leads to almost exactly the same results during the melting period. However, in the absence of wet snow (such as on March 5 in Fig. 7), $sign_{-2}$ and $sign_0$ performed better as they prevented over-detection of wet snow. For both individual and hybrid segmentations, initializing with the segmentation obtained for the previous date could be an interesting option, as it would account for some temporal continuity in the snow-cover development.

Our algorithm could be refined and extended in various ways. For instance, the information quality provided by optical sensors needs to be quantified. As the Chan–Vese energy minimization depends on the whole image, a global cloud cover percentage for the respective massif gives a first indication of the quality. However, the cloud localization also plays an essential role and may be considered in the creation of quality flags. Another question was raised with respect to the calculation of the SAE from the segmentation result, where the snowline altitude corresponds to the location of the first (wet) snow pixel in upward direction. Especially if the snow cover is interrupted vertically, this estimation needs to be questioned. Adapting the intensity of morphological filtering of the segmentations before the SAE calculation allows for some adjustments in this regard. Finally, to provide reliable hybrid SAEs, it needs to be considered that any meltline information from SAR acquisitions will influence the hybrid segmentations. This may be targeted by modification of the SAR image values in the orientation–altitude diagrams above the SAE, but the issue remains quite demanding. In a preliminary test for a date in April, the hybrid segmentation showed a meltline only when the weight of the SAR input was set to 0.9 or 1.

It could be argued that good results may also be achieved with different segmentation methods applied to the signal in orientation–altitude form. This is true and we stay open to other approaches. In the frame of the present work, we have benefited from the versatility of the method with respect to different data types, which provides a good foundation for data fusion. However, there may be cases where traditional thresholds could present a more reliable option for individual segmentations, since the method of Chan–Vese can over-detect (wet) snow if the initial condition or parameters are not well-tuned.

Lastly, we would like to emphasize the influence of orientation on SAEs, which is corroborated by different studies (see [50]). For example, Fig. 9 (Grandes Rousses, June 9, 2018) indicates SAEs around 2500 m for south-facing slopes and lower SAEs around 2300 m for north-facing slopes (both with SAR and optical data). This is due to the fact that differences in vegetation and solar radiation entail a significant variability in snow cover. However, for some specific situations, such as rain on snow events, SAEs can become similar at different orientations (e.g., Fig. 7, Vanoise, April 4, 2018). We should note also that variability in vegetation at lower elevations leads to important differences in detected snow by SAR.

VII. Conclusion and Outlook

We have introduced a novel approach to snowline altitude monitoring at the scale of mountain massifs, which consists in the application of an extended method of Chan–Vese to orientation–altitude diagrams displaying mean BSR values. Based on SAR data, our algorithm provides SAEs every six days, independently of cloud cover, which can be employed by end users such as the Météo-France snow and avalanche forecasters. We have seen that results compare well to independent data such as BERA and FSC for southern orientations, generally lying slightly above the reference products. For northern orientations, we observe a larger gap that remains to be investigated. As we have showcased with S2 data, the algorithm can be extended to a wide range of data types, since it does not fundamentally rely on the data-dependent

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thresholds often used in snow monitoring. Moreover, the orientation–altitude reference frame allows for the incorporation of products with different native spatial resolutions. This constitutes the foundation for hybrid SAEs, fusing optical and radar data, at smaller time intervals.

Rather than an exhaustive investigation of the data fusion, this study serves as a demonstration of its feasibility. We have chosen a snow melt period where dry snow is scarce. In scenarios where wet snow is the predominant snow, it is advantageous to integrate SAR and optical data: optical data may be missing in overcast regions, which can be investigated using SAR. SAR, on the other hand, is complex to interpret as it does not detect dry snow and is sensitive to other factors, such as soil moisture and vegetation cover. Combining the two types of data will allow us to make full use of the available remote sensing information. The fusion of SAR and optical data in situations where wet and dry snow both cover extended areas is more subtle, but presents a promising approach to identifying both types of snow. To tackle this problem, additional developments of our method are needed and will be tested in future studies.

In our proof of concept, we have assigned equal weight to each data source. In future studies, weights could be modulated according to cloud cover for optical data or vegetation types for SAR data. In addition, data fusion work may also involve data with significantly varying resolutions (e.g., VIIRS and MODIS), needing a more comprehensive examination of the weight to be ascribed to each data source and the ability to more effectively manage the difference in observation dates. To a certain extent, the hybrid product could also take into account the connection between consecutive dates, for instance through the initial condition for the segmentation.

The algorithm's performance will need to be tested for cases of poorer snow cover (as in the season 2021/22, or for massifs whith lower altitudes) and in different regions (such as the Pyrenees). Moreover, upcoming articles will present and evaluate our method's capacity for meltline detection (i.e., the limit between wet and dry snow) by means of SAR, which can be performed simultaneously to the snowline altitude estimation.

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